LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART IV

SESSION 16

Capstone & Integrative Review

From markets, to macro, to decisions that matter — in Nigeria and the world.



Reading: All sessions — micro to macro, domestic to global, policy to profits · Facilitator: Bongo Adi, Kelechi Nnamdi & Elvis Ozegbe

TODAY'S PLAN

Session 16 at a Glance — 75 Minutes

Connect micro to macro, domestic to global, policy to profits — and say it to a board in one page.



Capstone Presentations

Executive macro memos to a board, by industry

Integrative Review

Sixteen sessions on one page

Exam Preparation

Format, practice questions, common traps

WHERE WE ARE

Course Roadmap — 16 Sessions, 4 Parts

Pre-module and intra-module online sessions, sixteen on-campus sessions, and a closed-book final exam.

Part I



Sessions 1–5

Micro & markets

Data literacy, demand and elasticity, forecasting, market structure and platforms, market failures

Part II



Sessions 6–10

Macro, policy & risk

AD–AS and inflation, national accounts, monetary policy, fiscal policy and debt, financial stability

Part III



Sessions 11–12

Open economy

Exchange rates, capital flows and the trilemma; trade, supply chains, industrial policy and AfCFTA

YOU ARE HERE

Part IV



Sessions 13–16

Strategic frontiers

Climate economics, country risk and scenarios, hedging and resilience, capstone memos

BY THE END OF TODAY

Session 16 Learning Objectives

You will be able to:

01

Write a Board Memo

Turn macro evidence into a one-page recommendation a board can act on.

02

Connect the Course

Trace how a single policy or global shock flows through markets, prices, rates and the exchange rate to your P&L.

03

Prepare for the Exam

Know the format, practise the core reasoning, and avoid the common traps.

THE HOOK

The Board Has Ten Minutes

“Don't give me forty slides on the economy. Tell me what to do, why, and what would change your mind.”

That is the brief for your capstone. A good memo does three things:

Leads with the answer

A recommendation in the first sentence — price, invest, hedge, wait.

Shows the evidence

Three or four numbers that matter for this industry, with their direction.

Admits uncertainty

Scenarios, signposts and the trigger that would change the plan.

Your Capstone: A Macro Memo to the Board

The situation

Each team chooses an **industry** (cement, flour milling, banking, consumer goods, telecoms, power…) and acts as its strategy team.

The decision question

Given the economy today and the plausible paths for the next 12 months, **what should the board do** — on pricing, investment, financing and risk?

The data we have

Any course dataset plus official sources (CBN, NBS, IMF, World Bank, FRED). The **capstone starter script** builds a four-panel evidence pack. *Course data are stylised.*

What a good answer looks like

A **one-page memo** and an 8-minute presentation: recommendation, evidence, scenarios with probabilities, signposts and triggers, and owned actions.

PART

01

Writing the Memo

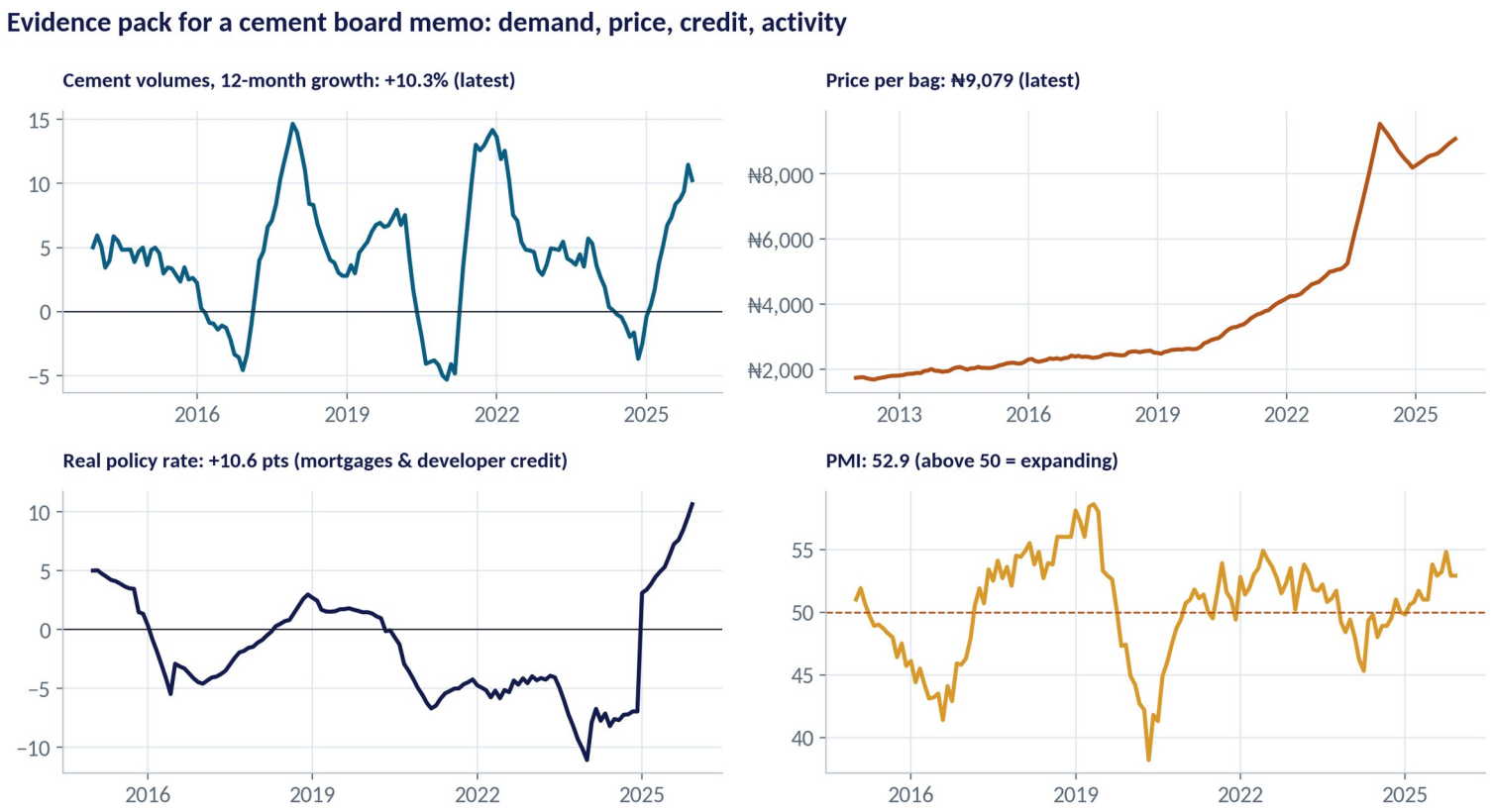
One page, one recommendation.

Six Paragraphs, One Page

Section	What to write	Course tools
1. Recommendation	One sentence: what the board should decide, and when	—
2. What has changed	Three or four macro numbers that matter to your industry, with direction	Dashboard (S1), inflation (S6), FX (S11)
3. What it means for us	Effect on prices, volumes, costs and financing	Elasticity (S2), rates (S8, S10)
4. Scenarios	Upside / base / downside, with probabilities and P&L impact	Scenarios (S14), forecasts (S3)
5. Signposts	Five indicators with trigger levels	Signpost board (S14)
6. Actions & owners	Pricing, hedging, capex, financing — who does what by when	Hedging (S15), climate (S13)

Write the recommendation last, then move it to the top.

Worked Example: The Cement Industry



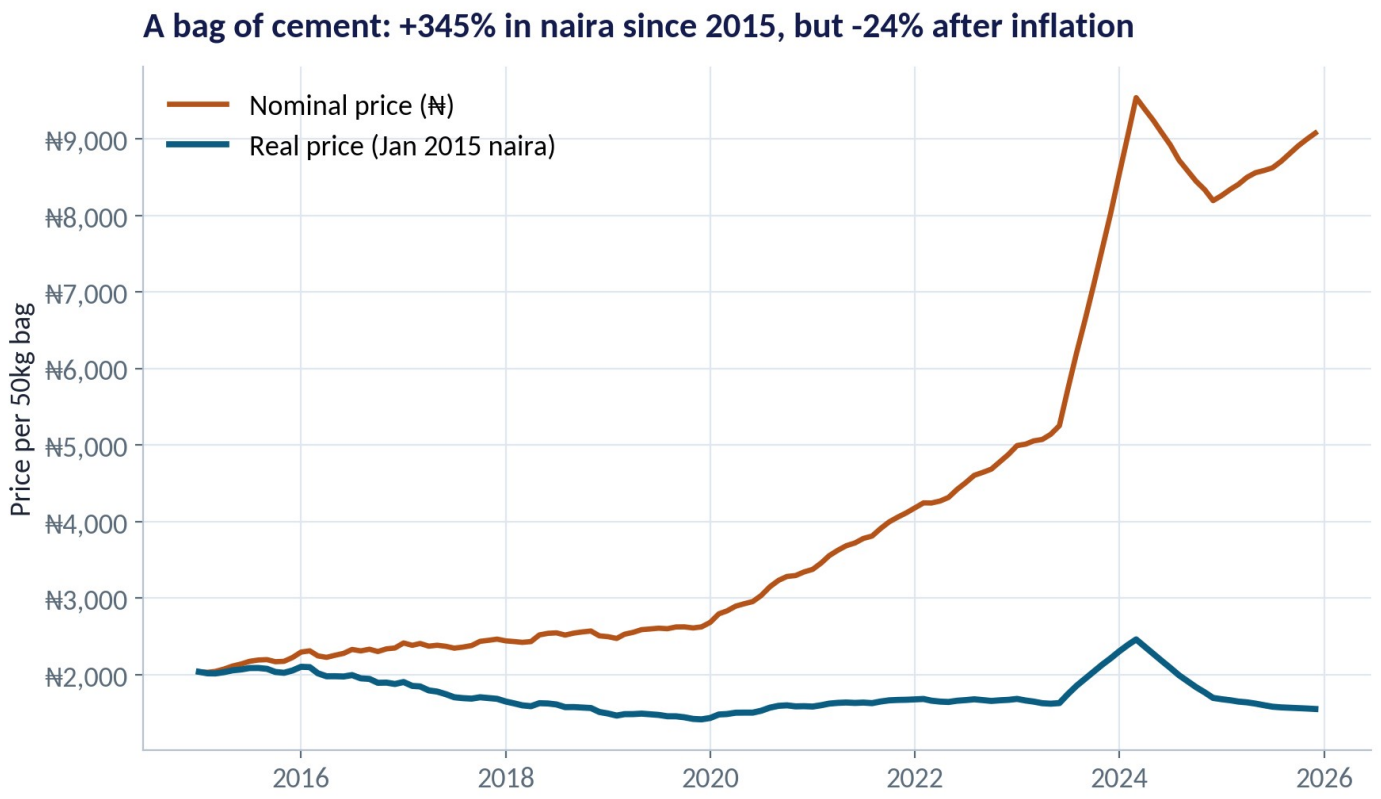
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S16-F04 · replicate: s16_figures.R / .py

WHAT TO SEE

- Volumes: **+10.3%** over the last 12 months.
- Price per bag: about **₦9,079**.
- Real policy rate: **+10.6 points** — expensive mortgages and developer credit.
- PMI **52.9**: activity expanding. Evidence for a cautious 'grow, but protect margin' memo.

Don't Be Fooled by Naira Prices



Real price uses an approximate CPI built from year-on-year inflation; the 2025 rebasing adds noise. Stylised data.

Fig S16-F05 · replicate: s16_figures.R / .py

WHAT TO SEE

- The price of a bag of cement rose **+345%** in naira since 2015.
- Adjusted for inflation, it **fell about 24%**.
- A memo that celebrates naira price growth without deflating it misleads the board.
- Session 1's first lesson is still the most important one.

WORKED EXAMPLE

A Cement Memo in Six Lines

Item	Evidence
Volumes	+10.3% y/y
Price/bag	₦9,079
Real rate	+10.6 pts
PMI	52.9
Naira	₦1,454/$

STEP BY STEP

1. **Recommend**: approve the second kiln line in two stages; stage one now, stage two on a trigger.
2. **Changed**: demand recovering, PMI above 50, naira stable for a year — but credit is costly.
3. **For us**: margins depend on energy costs in dollars and on price discipline in a concentrated market.
4. **Scenarios**: upside 30%, base 50%, downside 20% (naira above ₦2,000).
5. **Signposts**: PMI, 12-month FX change, reserves, permits; stage two if PMI > 52 for two quarters.

Actions: hedge 50% of next year's imported energy; finance stage one in naira at fixed rates; publish TCFD-style emissions data before the EU CBAM bites.

What Makes a Weak Memo

THE TRAP	THE BETTER HABIT
✗ Ten paragraphs of economic history before the recommendation.	✓ **Lead with the decision**; history only where it changes the decision.
✗ Naira growth numbers with no adjustment for inflation.	✓ Report **real** and **dollar** values alongside naira.
✗ One forecast presented as fact.	✓ **Three scenarios** with probabilities and a downside you can survive.
✗ Actions without owners or dates.	✓ **Who, what, by when** — and the trigger that brings stage two forward.
✗ Ignoring the global picture.	✓ Show how the **Fed, oil and trade** reach your industry.

PART

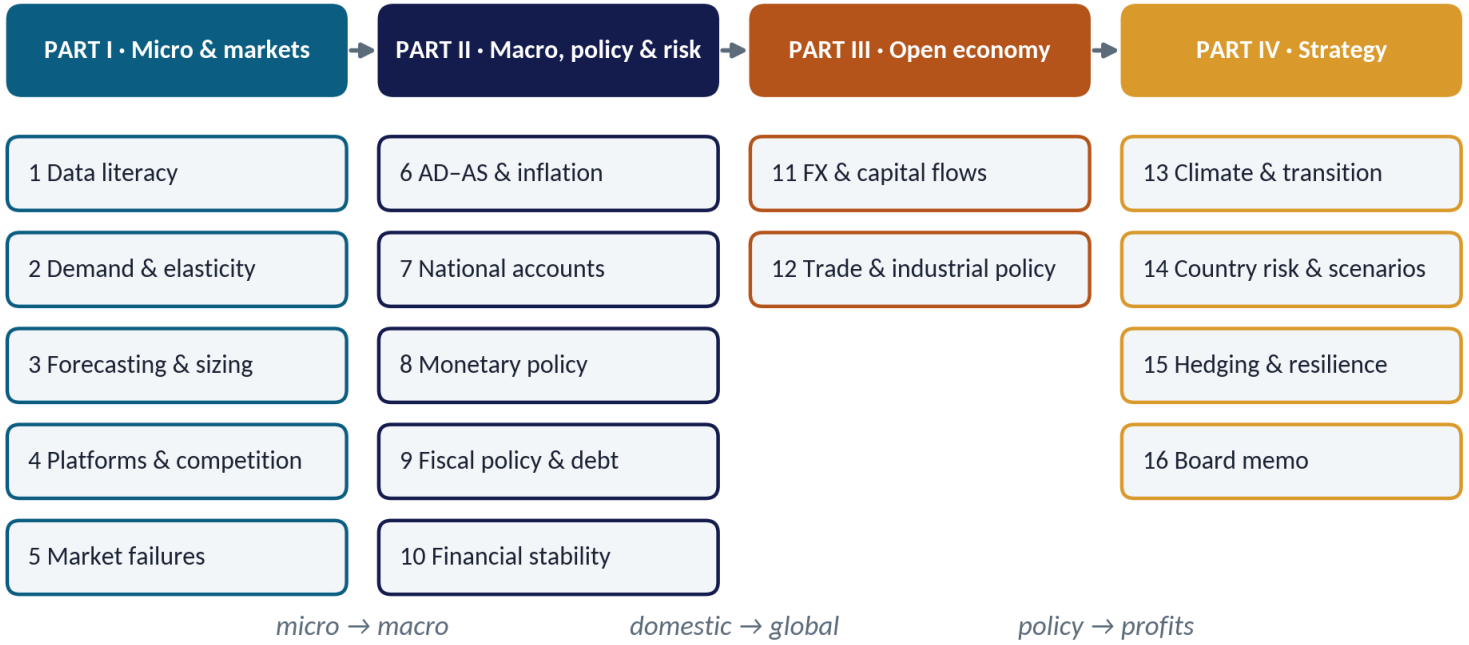
02

The Course on One Page

Micro to macro, domestic to global, policy to profits.

Sixteen Sessions, One Logic

Sixteen sessions, one argument



Every board memo travels the whole map: a market question, the macro forces on it, the global links, and a decision with a hedge.

Fig S16-F02 · replicate: s16_figures.R / .py

WHAT TO SEE

- **Part I**: how markets set prices and quantities, and where they fail.
- **Part II**: how the whole economy moves — inflation, output, money, budgets, banks.
- **Part III**: how Nigeria connects to the world — currency, capital and trade.
- **Part IV**: how a firm turns all of it into strategy — climate, country risk, hedging, the board memo.

One Shock, Four Panels

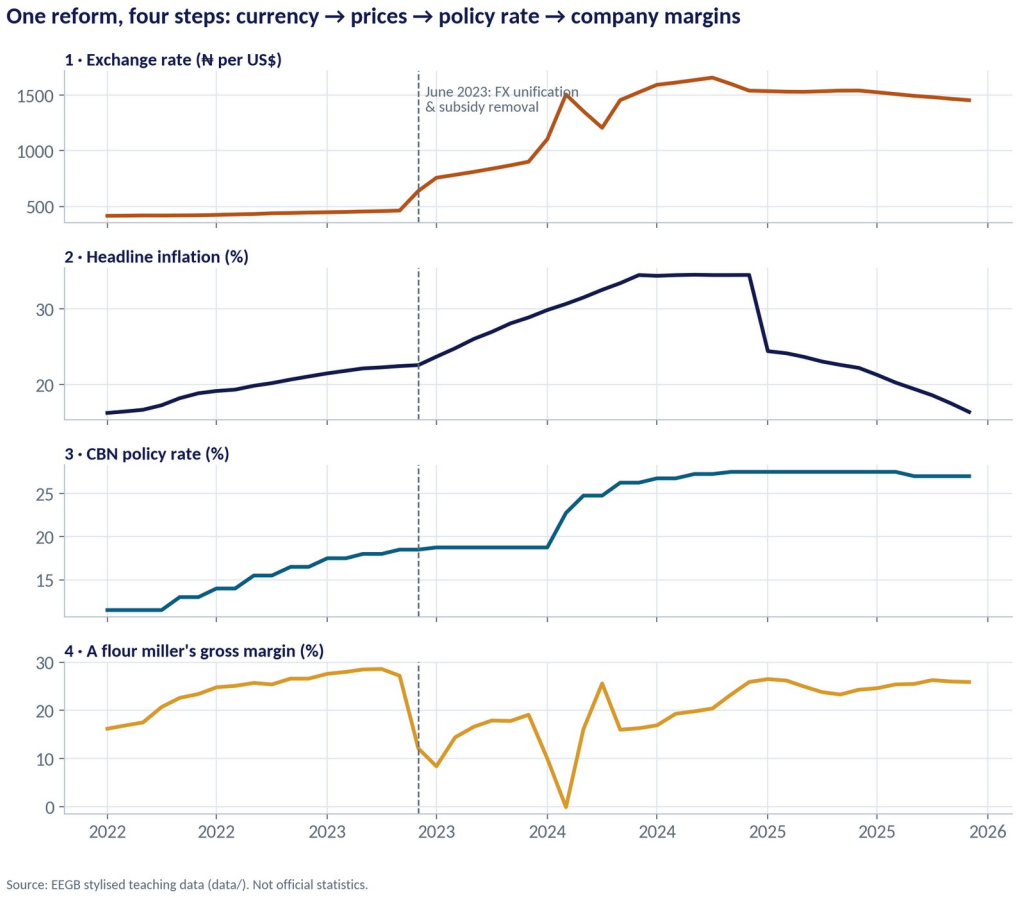
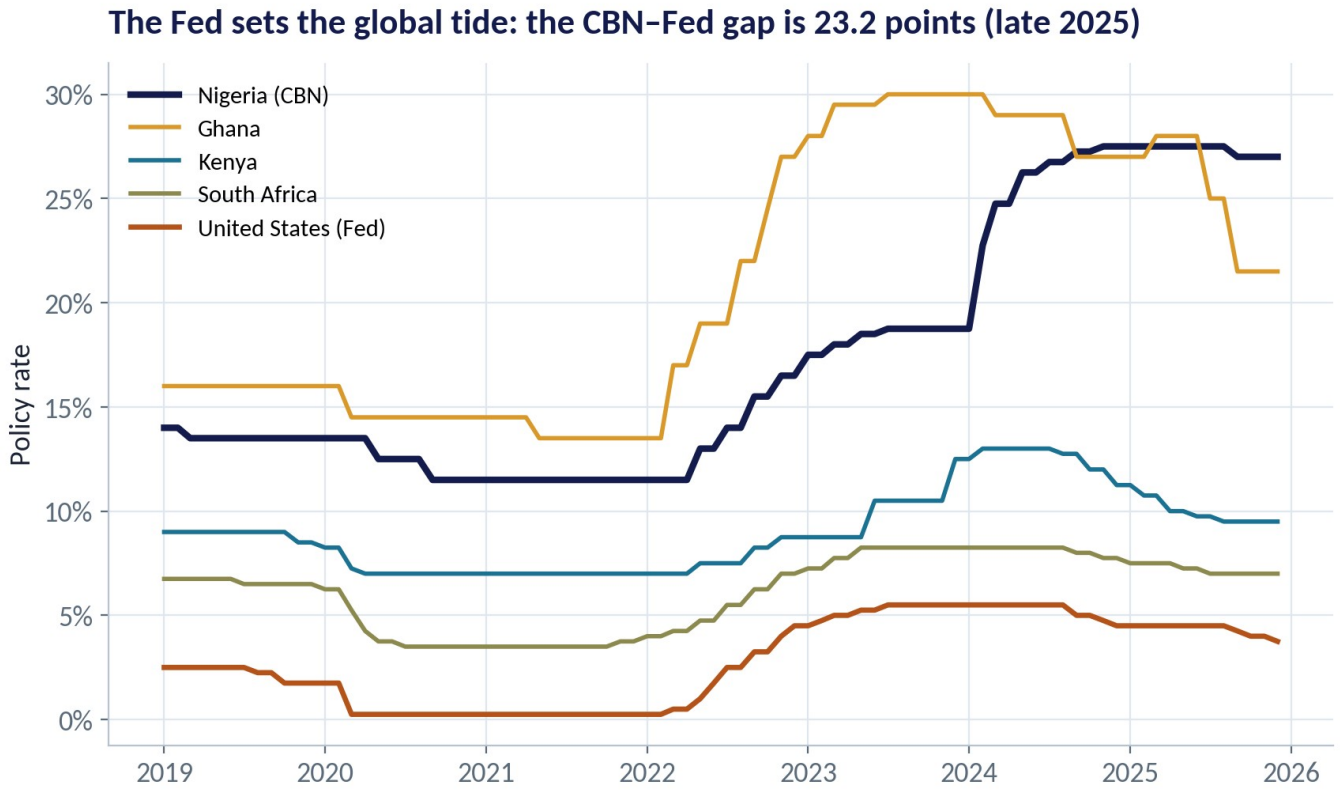


Fig S16-F01 · replicate: s16_figures.R / .py

WHAT TO SEE

- June 2023: FX unification and petrol subsidy removal.
- The naira moved from about **₦463** to a peak near **₦1,657** per dollar.
- Inflation peaked near **34.5%**; the CBN raised its rate from **11.5%** (early 2022) to **27.5%**.
- A flour miller's gross margin fell to about **zero** at the worst point — policy reaches the P&L within months.

The Global Tide and Nigeria's Rate



Source: EEGB stylised teaching data (data/). Not official statistics.

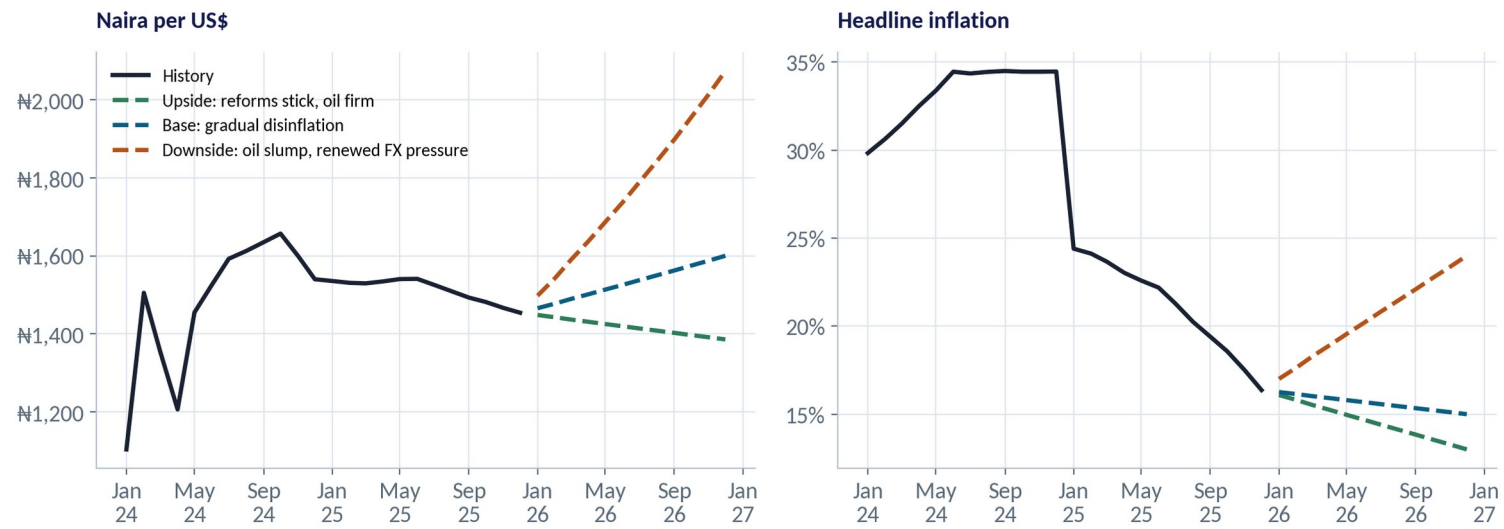
Fig S16-F06 · replicate: s16_figures.R / .py

WHAT TO SEE

- The US Fed's rate moves capital worldwide; emerging markets must pay a premium to keep it.
- In late 2025 the CBN's rate was about **23 points** above the Fed's.
- Part of that gap is Nigerian inflation; part is the risk premium investors demand.
- When the Fed cuts, pressure on the naira usually eases — one signpost for your memo.

Three Scenarios for 2026

Write the memo for three futures, not one forecast



Scenario paths are illustrative assumptions starting from the last data point, not forecasts.

Fig S16-F03 · replicate: s16_figures.R / .py

WHAT TO SEE

- **Upside** (reforms stick, oil firm): naira about **₦1,386**, inflation about **13%** by end-2026.
- **Base** (gradual disinflation): about **₦1,600** and **15%**.
- **Downside** (oil slump, renewed FX pressure): about **₦2,073** and **24%**.
- These are teaching scenarios, not forecasts — build your own for the memo.

Five Chains to Know for the Exam

Start	Chain	Reaches the firm as…
Oil price falls	Fewer dollars → reserves fall → naira weakens → imported inflation	Higher input costs, weaker demand
CBN raises rates	Higher yields → dearer credit → slower demand; bond prices fall	Costlier loans, lower valuations
Fiscal deficit financed by CBN	More money → inflation → weaker naira	Rising costs, policy uncertainty
Fed raises rates	Capital leaves EMs → FX pressure → CBN tightens	FX and rate risk at once
Carbon price / CBAM	Emitters pay → prices rise or margins fall → capital moves to clean firms	Transition risk and opportunity

PART

03

Preparing for the Final Exam

Closed-book, in class: reasoning over memorising.

THE FORMAT

What to Expect

❐ Closed-book, in class

On-site, no notes or devices. It counts for **40%** of your grade.

✦ Reasoning, not recall

Expect short cases: a shock, a firm, a question. Explain the chain and recommend.

± Simple numbers

Real vs nominal, elasticity and revenue, the debt equation, duration, the cost of a dollar loan.

✎ Clear writing

Short paragraphs, the answer first, the reasoning after. Draw a diagram when it helps.

Try These Before the Exam

#	Question	What a good answer covers
1	A firm's naira revenue rose 25% while inflation was 30%. Did it grow?	Real ≈ −5%; volume fell; check prices vs competitors
2	Demand elasticity is −1.5. Should we raise prices 10% to pass on costs?	Volume −15%, revenue falls; margin vs volume trade-off
3	Why can't the CBN fix the naira, allow free capital flows and cut rates?	Impossible trinity; reserves drain; parallel market
4	r = 18%, g = 12%, primary deficit 1% of GDP, debt 50%. What happens to debt/GDP?	Rises: (r−g)/(1+g)×d ≈ 2.7 pts + 1 pt deficit
5	A bank's duration gap is 2 years. What does a +3 point rate rise do?	Equity falls ≈ gap × assets × 3% ÷ (1+y)
6	When is a petrol subsidy regressive?	Benefits rise with consumption; richer deciles gain most

Quick Check: What Would You Say?

?

Practice: the Fed cuts rates by 1 point while Nigerian inflation is stable. Which chain is most likely for a Lagos importer?

A. Higher naira borrowing costs immediately

B. Capital flows toward emerging markets, easing pressure on the naira and the cost of imported inputs

C. No effect, because the Fed only affects the US

D. Higher Nigerian inflation

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

Quick Check: What Would You Say?

?

Practice: a board member says 'our dollar loan at 7% is cheaper than a naira loan at 22%'. What do you check first?

A. Nothing — 7% is lower

B. The expected depreciation of the naira and whether our revenues are in naira (currency mismatch)

C. The Fed's balance sheet

D. The CPI rebasing

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Capstone Starter: Your Evidence Pack

YOU WILL NEED

Data

`data/ng_macro_monthly.csv`
`+ one industry file`

R script

`sessions/s16/R/`
`s16_capstone_starter.R`

Python script

`sessions/s16/python/`
`s16_capstone_starter.py`

Before your presentation

1. Set **INDUSTRY** and choose the **four macro drivers** that matter most to it.
2. Run the script: it prints an **evidence table** (latest, a year ago, change).
3. It saves a **four-panel chart** for your memo.
4. Add one **industry dataset** or official series.
5. Fill in the **six-paragraph memo** template it prints — then rewrite in your own words.

Deliverable: a one-page memo and an 8-minute presentation to the board.

The Words That Carry the Course

Real vs nominal	Adjusted vs not adjusted for inflation (S1).
Elasticity	% change in quantity for a 1% change in price or income (S2).
Externality	A cost or benefit falling on others (S5, S13).
Supply shock	A cost shock that raises prices and cuts output (S6).
Transmission	How policy reaches firms and households (S8).
r – g	Interest rate minus growth: drives the debt ratio (S9).
Duration gap	How a rate rise hits a balance sheet (S10).
Impossible trinity	Fixed rate, free capital, own policy: pick two (S11).
Scenario & signpost	Plausible futures and the indicators that reveal them (S14).
Natural hedge	Matching costs and revenues in the same currency (S15).

KEY TAKEAWAYS

EEGB in Five Ideas

01 Think in **incentives, margins and opportunity cost** — and always separate real from nominal.

02 Prices, inflation, rates and the naira are **linked**: one shock travels through all of them.

03 Nigeria is an **open economy**: oil, the Fed and capital flows shape what the CBN can do.

04 Plan with **scenarios, signposts and hedges**, not a single forecast.

05 Tell the board **what to do**, why, and what would change your mind.



NEXT › Final exam

Next: Submit your one-page memo before your presentation slot. Revise the practice questions and the five transmission chains for the closed-book final exam.